Improved three-point estimation method for probabilistic power flow calculation

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***Abstract*—The large-scale grid connection of new energy power generation has brought uncertainty to the power system due to its own intermittency. At present, the three-point estimation method is one of the commonly used methods of probabilistic power flow calculation for power system planning, operation and control of uncertain quantities. However, this method has the problem of insufficient tail accuracy. For this reason, this paper proposes to add a set of normal distribution tail points and combine the three-point estimation method to calculate, use Nataf transform and Cholesky decomposition to process the correlation input variables, and use Cornish- Fisher series fit to get the probability distribution function. The system of IEEE14 node, IEEE30 node and IEEE57 node was tested, and the results verified the correctness of the improved three-point estimation method of probability power flow algorithm.**

***Keywords*—I3PEM; correlation; Nataf transform; Cornish-Fisher series; new energy generation**

I. Introduction

The development of distributed power sources such as wind power and photovoltaic can promote society's rapid progress towards a low-carbon and high-efficiency society [1]. Wind speed and light intensity will be affected by factors such as climate, cloud cover and geographical location, and there is often a correlation[2], and Its large-scale grid connection will have a significant impact on the safety and stability of power system operation. Traditional deterministic power flow calculations cannot solve such problems. Therefore, it is particularly important to conduct probabilistic power flow calculation research for this uncertainty. At present, there are three main methods for probabilistic power flow research: PEM[3] (Point Estimate Method), analysis Method[4], MCS[5] (Monte Carlo Simulation). Among them PEM is an approximate algorithm that uses the digital characteristics of the input random variables to approximate the statistical characteristics of the system state variables. Compared with the analytical method, it is simpler and easier to handle. Compared with the analog method, it has the advantage of speed. Therefore, it is widely used in the probabilistic power flow calculation.

PEM can obtain the digital characteristics of the output variable through calculation when the input random variable is known, and obtain the estimated value of each order moment. But it cannot get the PDF of the output variable, so it is necessary to combine the series expansion method to obtain the probability density function PDF (Probability Density Function) and the cumulative distribution function CDF (Cumulative Distribution Function) method. The traditional point estimation method has 2PEM (Two-Points Estimation Method) and 3PEM (Three-Point Estimation Method). The latter is more accurate than the former because of the increase in the number of estimated points. Reference[6], on the basis of the point estimation method, a pair of uniform probability estimation points are added for improvement. Reference[7] uses the commonly used correlation processing method in the analytical method, Nataf transformation method to process non-normal input, and combines 3PEM to calculate and solve. As the number of points increases, the accuracy will increase, but the time consumption also increases[8]. In the literature[9-10], in the wind-solar hybrid system, it is proposed to use the wind-solar output power ratio as an influencing factor to redistribute the weight of each point to improve the accuracy of the point estimation method. The above-mentioned documents are all improving the two-point estimation method or dealing with the correlation, while for the three-point estimation method, there are few literatures to improve the method.

In view of the lack of improvement of the three-point estimation method, the article proposes an improved three-point estimation method I3PEM (Improved Three Point Estimate Method), which adds a set of normal distribution tail points for additional calculations, and then combines the three-point estimation method that improve the calculation accuracy

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of the three-point estimation method. The correctness of the proposed algorithm is verified by IEEE14 , IEEE30 and IEEE57 systems.

II. Point Estimate Method

Let F be the functional relationship between the input random variable $X=[x_1, x_2,..,x_n]^T$ and the output random variable H, as shown in the following formula.

$$H = F(X) = F(x_1, x_2, ..., x_n) \quad (1)$$

The point estimation method is to find m points in each input variable to represent the input value of this random variable. The formula is as follows:

$$x_{i,k} = \mu_{x,i} + \xi_{xi,k}\delta_{xi} \quad k=1, 2, \dots m \quad (2)$$

In the formula, μ_{si}, σ_{xi}, ξ_{xi} are expectation, standard deviation, and position coefficient.

Set $P_{xk,i}$ as the weight coefficient of the sampling point $X_{k,j}$, the formula is

$$\begin{cases} \sum_{i=1}^{m} p_{xk,i} = \frac{1}{n} \\ \sum_{k=1}^{n}\sum_{i=1}^{m} p_{xk,i} \end{cases} \quad (3)$$

$\xi_{xi,k}$, $P_{xk,i}$ can be obtained by the following formula:

$$\sum_{k=1}^{m} p_{xi,k} \cdot \left(\xi_{xi,k}\right)^j = \lambda_{i,j}, j = 1,2,\dots,2m-1, i = 1,2,\dots n \quad (4)$$

$\lambda_{i,j}$ is shown in the following formula

$$\begin{cases} M_j(X_i) = \int_{-\infty}^{+\infty} (x-\mu_i)^j f(x)dx, i = 1,2,...,n \\ \lambda_{i,j} = \frac{\beta_j(X_i)}{(\sigma_i)^j} = E\left(\frac{X_I-\mu_i}{\sigma_i}\right)^j, j = 1,2,3... \end{cases} \quad (5)$$

It can be seen from the above formula that it is the central moment, which is the standardized central moment. The 1-4 order standardized central moments of the normal distribution are 0, 1, 0, 3. Among them, the third-order standardized central moment is called skewness, and the fourth-order standardized central moment is called peak value.

After obtaining the estimated value of each order of H:

$$E(H^J) \approx \sum_{i=1}^{n}\sum_{k=1}^{m} P_{xi,k}\left[F\left(\mu_1, \mu_2, ..., x_{k,i}, ..., \mu_n\right)\right]^j \quad (6)$$

Generally, the more points are estimated, the more accurate the estimated value can be obtained. However, it can be seen from formula (5) that the higher the order, the more difficult it is to obtain.

A. Three-Point Estimation Method

When m=3, it is called 3PEM. The formula for obtaining the position coefficient is as follows:

$$\xi_{xi,k} = -\frac{\lambda_{i,3}}{2} + (-1)^{3-k}\sqrt{\lambda_{i,4} + \frac{3}{4}\lambda_{i,3}^2} \quad (7)$$

The weight coefficient $P_{xi,k}$ is

$$\begin{cases} P_{xi,k} = \frac{(-1)^{3-k}}{\xi_{xi,k}(\xi_{xi,1}-\xi_{xi,2})} \\ P_{xi,3} = \frac{1}{m} - P_{xk,1} - P_{xk,2} = \frac{1}{m} - \frac{1}{\lambda_{i,4}-\lambda_{i,3}^2} \end{cases} \quad (8)$$

According to the skewness and kurtosis being 0 and 3 respectively, $\xi_{i,1}$, $\xi_{i,2}$, $\xi_{i,3}$ it can be obtained. Substituting into equation (6), the estimated moments of each order of H can be obtained.

B. Improved Three Point Estimate Method

In 3PEM, the skewness and kurtosis of the standard normal distribution are 0 and 3, respectively. According to formulas (7) and (8) $\xi_{xk,1}$, $\xi_{xk,2}$, $\xi_{xk,3}$ is $\sqrt{3}$, $-\sqrt{3}$,0 can be obtained. However, there are no other sampling points in the normal space for calculation. According to the principle of increasing the number of points to improve accuracy, it is proposed to add a set of normal distribution tail points X_i' because of the three-point estimation method to improve the tail characteristics. From equation (9), the sampling points of the two-point estimation method of the standard normal distribution are usually at the tail of the normal distribution, so the newly added position coefficient and the weight of each point are as follows:

$$\begin{cases} \xi_{xi,1} = \sqrt{n}, \xi_{xi,2} = -\sqrt{n} \\ P_{i,k} = \frac{1}{2n} \end{cases} \quad (9)$$

According to the principle of probability statistics, the normal distribution obeys 3 σ principles, that is, the probability of a point falling at 1 standard deviation from the mean is 68.2%, the probability of a point within 2 standard deviations is 95.5%, and the probability of a point at 3 times the standard deviation is 99.7%. The newly added tail points are often farther away from the average point than the point obtained by the three-point estimation method, so the estimated total weight of the newly added set of $2n-1/2n$ points should not be too large. This article sets the weight of the three-point estimation as the weight of the newly $1/2n$ added point. The formula (6) is rewritten as follows

$$E(H^j) \approx \left(\frac{2n-1}{2n}\sum_{i-1}^{n}\sum_{k=1}^{m} P_{xi,k} \times \left[F\left(\mu_1, ..., x_{k,i}, ..., \mu_n\right)\right]^j + \frac{1}{2n}\sum_{i-1}^{n}\sum_{k=1}^{m} \times \left[F\left(\mu_1, ..., x_{k,i}, ..., \mu_n\right)\right]^j\right) \quad (10)$$

It can be seen from the above formula that compared with the 2m+1 point estimation method; this method seems to be nearly doubled compared with the three-point estimation method. But because the position coefficients and weights of the newly added set of points are all given, the actual calculation increase is acceptable.

III. Modeling uncertain variables with correlation

A. Probability Model of Uncertain Variables

1) Wind power generation probability model

The wind speed PDF obeys the Weibull distribution[12], and its expression is:

$$f(v = \frac{k}{c}\left(\frac{v}{c}\right)^{k-1} \exp\left[-(\frac{v}{c})^k\right]$$

(11)

v, k, c is wind speed, shape parameter and scale parameter, The relationship of the active power function of the wind turbine is as follows:

$$P_{wg} = \begin{cases} 0, v \le v_{in} or v > v_{out} \\ P_r\frac{v-v_{in}}{v_r-v_{in}}, v_{in} < v \le v_r \\ P_r, \ v_r < v \le v_{out} \end{cases} \quad (12)$$

V_{in}, v_r, v_{oout}, are the minimum, rated, and maximum wind speeds respectively. Pwg, Pr are actual and rated power respectively.

2) Load probability model

Load usually exhibits uncertain characteristics, and normal distribution can generally be used to express this characteristic of load[13]. Its PDF is shown below.

$$\begin{cases} f(P_L) = \frac{1}{\sqrt{2P_L}\delta_p}\exp\left(-\frac{(P_L-\mu_p)^2}{2\delta_p^2}\right) \\ f(Q_L) = \frac{1}{\sqrt{2Q_L}\delta_Q}\exp\left(-\frac{(Q_L-\mu_Q)^2}{2\delta_Q^2}\right) \end{cases} \tag{13}$$

Where: P_L and Q_L are the active power and reactive power respectively;

μ_p and σ_p is the mean value and mean square deviation of active power respectively; μ_Q, σ_Q is the mean value and mean square deviation of reactive power respectively.

B. Correlation processing

There is often a certain correlation between wind power and load, and ignoring these correlations will lead to results that are inconsistent with the actual situation. Therefore, the correlation needs to be considered in the calculation. Nataf transformation is a transformation process that maps correlated non-normal random variables samples to independent standard normal random variables.

Suppose an n-dimensional non-normal input variable $X=[x_1,x_2...,x_n]$ with correlation C_x is, its correlation coefficient matrix ρ_{ij} is, and the element at the corresponding position is. Suppose the independent standard normal distribution variable $Y=[y_1,...y_n]$ is, its correlation coefficient matrix C_Y is, and the corresponding location element ρ_{oij} is.

The conversion relationship between X and Y is

$$\begin{cases} \Phi(y_i) = \mathrm{F}(x_i) \\ y_i = \Phi^{-1}(F_i(x_i)) \end{cases} \tag{14}$$

Among them, $F(.)$, $\Phi^{-1}(.)$ are the cumulative distribution function of X and the inverse cumulative distribution function of the standard normal distribution.

Where the correlation coefficient ρij is defined as

$$\rho_{ij} = \frac{cov(x_i x_j)}{\sigma_i\sigma_j} \tag{15}$$

σj is x_i, x_j the mean square error of. between ρ_{ij} and ρ_{oij} meet:

$$\begin{aligned} \rho_{ij} &= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\frac{x_i-\mu_i}{\sigma_i}\cdot\frac{x_j-\mu_j}{\sigma_j}f_{ij}(x_i,x_j)dx_idx_j \\ &= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}\frac{F_i^{-1}(\Phi(y_i))-\mu_i}{\sigma_i}\cdot\frac{F_j^{-1}(\Phi(y_j))-\mu_j}{\sigma_j}\cdot \\ &\quad \Phi_2(y_1,y_2,\rho_{ij})dy_idy_j \end{aligned} \tag{16}$$

However, the calculation of this formula is complicated, and the transformation of wind power correlation coefficient can be calculated by the empirical formula in literature[14].

The C_Y calculated Cholesky decomposition is performed $C_Y=LL^T$, and the lower triangular matrix L is obtained. By $Z=L^{-1}Y$ obtaining independent standard normal distribution random variables.

IV. Algorithm Flow

The power flow calculation equation can be expressed as equation (17)

$$H(U,\delta...) = F(P,Q) \tag{17}$$

In the formula, P and Q are the active and reactive power of the system nodes, and U, δ are the node voltage and phase angle, respectively.

After obtaining the moments of the origin of each order of the output variable, use Cornish-Fisher series to expand to obtain PDF and CDF. Set α as the quantile of a random variable, it $Z(\alpha)$ can be expressed as

$$\begin{aligned} Z(\alpha) &= \xi(\alpha) + \frac{\xi^2(\alpha)-1}{6}g_3 + \frac{\xi^3(\alpha)-3\xi(\alpha)}{24}g_4 - \\ &\frac{2\xi^3(\alpha)-5\xi(\alpha)}{36}g_3^2 + \frac{\xi^4(\alpha)-6\xi^2(\alpha)+3}{120}g_5 + ... \end{aligned} \tag{18}$$

Where, $\xi(\alpha)=\Phi^{-1}(\alpha)$

From $Z(\alpha)=F^{-1}(\alpha)$ this, the Z cumulative distribution can be obtained $F(z)$.

In order to verify the accuracy of the algorithm, the results obtained by the Monte Carlo simulation method considering the correlation are used as a comparison. Evaluate with the error index of formula (19) and formula (20).

$$\bar{\varepsilon_\mu} = \frac{\sum_{i=1}^{N_r}|\mu_{MCS}-\mu_{PEM}|/|\mu_{MCS}|}{N_r}\times 100\% \tag{19}$$

$$\bar{\varepsilon_\sigma} = \frac{\sum_{i=1}^{N_r}|\sigma_{MCS}-\sigma_{PEM}|/|\sigma_{MCS}|}{N_r}\times 100\% \tag{20}$$

μ_{MCS}, σ_{MCS} is the expectation and mean square deviation of the output variables obtained by the MCS, is the expectation and mean square deviation of the output variables obtained μ_{PEM}, σ_{PEM} by the point estimation method, and Nr is the number of output variables.The algorithm flow is shown in Fig.1.

V. Example Analysis

A. Example 1

It is verified with an improved IEEE14 system with a reference power of 100MVA. This system has 14 nodes and 13 branches. The total active load of the system is 259MW. Taking the load active power of the standard calculation example as the expectation, the system reactive load is simplified, that is, the node power factor is 0.75 and the correlation coefficient between loads is 0.8. Two sets of small wind farms with a total installed capacity of 20MW and 15MW are added to the system, and the units are operated in a constant power factor of 1. The shape parameter is 7.0, the scale parameter is 1.8, and the cut-in, rated, and cut-out wind speeds are 4m/s, 15m/s, and 25m/s, respectively. The improved IEEE14 system is shown in Figure 2. Two wind turbines are connected to node 11 and node 13, respectively, with a correlation coefficient of 0.7. At the same time, a group of photovoltaic electric fields are connected to node 14 with an installed capacity of 5 MW and a reactive output of 0. The shape parameters a and b are 0.6798 and 1.7888 respectively, the photoelectric conversion rate is 0.13, and the maximum light intensity is 1.1333kW/m2.

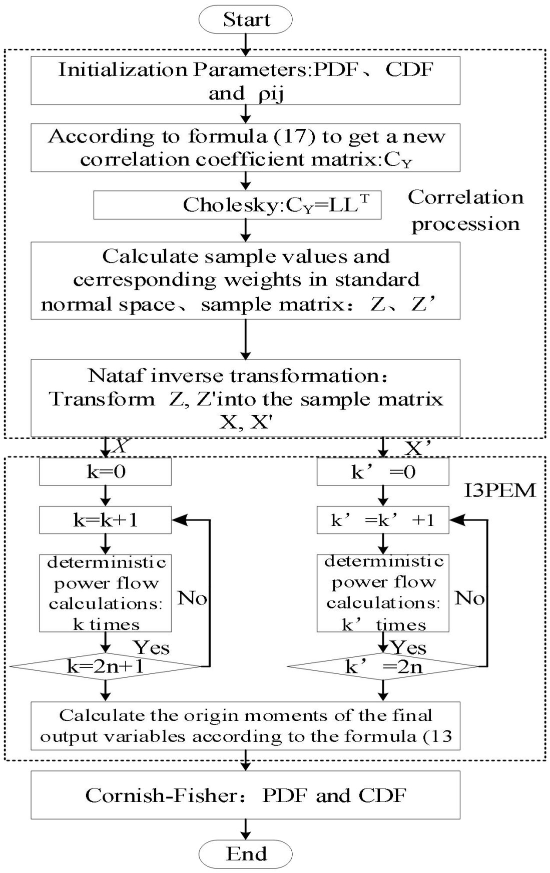


Fig.1 Process of improved three-point estimate method considering correlation

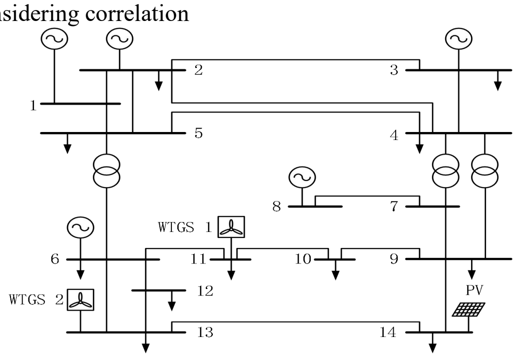


Fig.2 Wiring diagram of IEEE-14 bus system

The proposed I3PEM is used to calculate this improved IEEE 14 system, and the 10,000-time Monte Carlo simulation method is used as a reference to compare with 3PEM. The number of random input variables in this system n is 14, the correlation coefficient between the various loads of the system is set to 0.8, the load variation coefficient (the ratio of the standard deviation of the random variable to the absolute value of the expected value) is 0.05, and the calculation time of the three algorithms, such as shown in Table 1, the calculated error list is shown in Table 2.

As can be seen from Table 2, the overall mean error is reduced by 31.04%, and the standard deviation error is reduced by 37.83%. It can be seen from Table 1 that the calculation time of the three-point estimation method and the improved three-point estimation method are much shorter than the Monte Carlo simulation method, and the I3PEM calculation time is slightly longer than the 3PEM calculation time, but it is also within 0.5 seconds. In summary, in this scenario, I3PEM can maintain high work efficiency while considering accuracy.

The following scenarios are given to further study the performance of I3PEM:

Scenario A: The correlation coefficient between wind farms remains unchanged at 0.7, the load variation coefficient is 0.1, and the correlation between loads is set to (0.1, 0.3, 0.5, 0.7, 0.9).

Scenario B: The correlation coefficient between wind farms remains unchanged at 0.7, the correlation coefficient between loads is set to 0.8, and the load variation coefficient is set to (0.05, 0.1, 0.15, 0.2, 0.25, 0.3)

TABLE I. CALCULATION TIME OF DIFFERENT ALGORITHMS

method	3PEM	I3PEM	MCS
time/s	0.3031	0.4684	60.4634

TABLE II. AVERAGE RELATIVE ERROR

idnex	method	Index value/%				
		V	δ	Pij	Qij	Ploss
$\bar{\varepsilon}_{\mu}$	3PEM	0.0014	0.1430	0.4687	0.4685	0.1504
	I3PEM	0.0013	0.0879	0.3322	0.3320	0.0962
$\bar{\varepsilon}_{\sigma}$	3PEM	0.7866	0.6468	1.3439	0.8104	1.2720
	I3PEM	0.6123	0.1555	0.8574	0.4575	0.9981

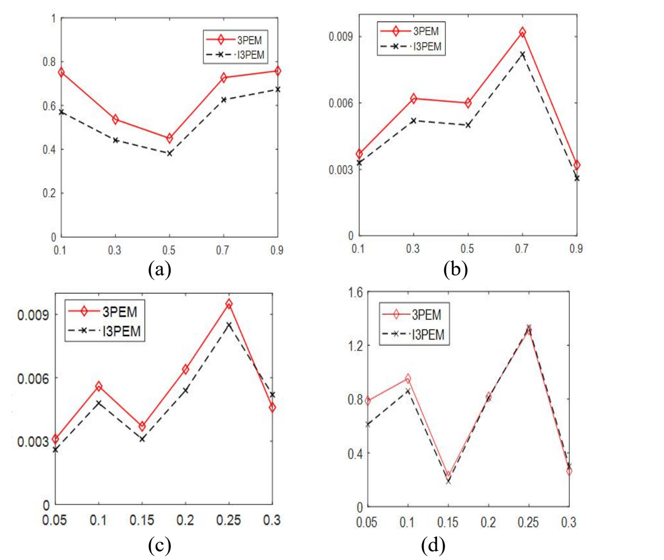


Fig.3 IEEE14 node: different correlation coefficients, different coefficients of variation $\bar{\varepsilon}\mu$ and $\bar{\varepsilon}\sigma$

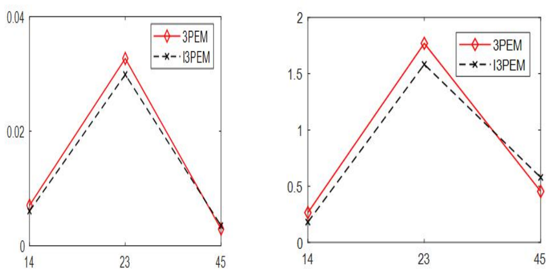
Fig.3 a and b are in scenario A. Under different correlation coefficients, the I3PEM error is less than the 3PEM error, and the error value reduction is more than 20%. Therefore, I3PEM is calculated under different correlation coefficients. The result is more accurate. Fig.4 c and d show scenario B. As can be seen from the figure, within the range of the load variation coefficient of 0.05~0.2, the expected value of the voltage amplitude and the standard deviation error of I3PEM are both less than 3PEM, so I3PEM is more It is suitable for the load variation coefficient of 0.2 and below. At the same time, the smaller the variation coefficient, the more excellent the I3PEM can be reflected, and it is also more suitable to use I3PEM.

B. Example 2

In order to compare the influence of the number of random input variables on I3PEM, this calculation example performs probabilistic power flow calculations on IEEE 14 system, IEEE 30 system, and IEEE 57 system. The active power of the system load is the standard case active power, and the reactive power is calculated with a constant power factor of 0.75, and other parameters remain unchanged. Wind farm access node number, photovoltaic access node number, load node number, and the number of random input variables in each system (number of random input variables: number of wind power + number of photovoltaic + number of load nodes) are shown in Table 3. Wind power the specific parameters of field and photovoltaic are the same as in Example 1.

TABLE III. NUMBER OF RANDOM INPUT VARIABLES IN DIFFERENT SYSTEM

system	Wind farm access node	PV access node	Number of load nodes	random input variables n
IEEE14	11,13	14	11	14
IEEE30	16,19	21	20	23
IEEE57	37,44	17	42	45



(a) (b)

Fig.4 IEEE30 and IEEE57 node: Number of different input variables $\overline{\varepsilon\mu}$ and $\overline{\varepsilon\sigma}$

Fig.4 (a) and (b) are respectively the expected value error index and standard deviation error index of different system I3PEM when the system load variation coefficient is 0.1. It can be seen from the figure that when the number of random input variables exceeds 23 hours, the expected error and standard deviation error of the improved three-point estimation method are smaller than those of the three-point estimation method.

VI. CONCLUSION

Due to the three-point estimation method, when calculating the probabilistic power flow problem caused by new energy grid connection, there is insufficient tail accuracy. The article proposes an improved three-point estimation method I3PEM, which adds a set of normal distribution tail points for additional calculations, and then combines the three-point estimation method to improve the calculation accuracy of the three-point estimation method . The Nataf transformation method and Cholesky decomposition are used to obtain the random input variables of wind power and load considering the correlation, and the improved algorithm is used to solve the probabilistic power flow. The correctness of the proposed algorithm is verified by the IEEE14 and IEEE30, IEEE57 node system.

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